

Guidance for Writing the Final Project Manuscript

All teams (or individuals) will submit an 8–10 page manuscript that mirrors the structure and rigor of a publishable paper.

A. Overall Expectations

The manuscript should tell a clear, logical, and quantitative story.

Write as if you were reporting your work to a professional scientific audience.

Clarity, reproducibility, and rigor are more important than length.

Use consistent formatting, proper grammar, and visual clarity in figures/tables.

Formatting

Length: 8–10 pages (excluding appendices and code)

Font: 12-pt, 1.0 line spacing

Figures and tables numbered and captioned

Codes and raw data are provided in an appendix at the end of the manuscript and are excluded from the page count.

B. Recommended Manuscript Structure

Title Page

Title: Concise and descriptive of the design and study question.

Author(s): Include all team members, majors, departments, and contribution statements.

Abstract (≤250 words): Summarize the objective, design, analysis, and main findings.

1. Introduction

Introduce the scientific or applied problem motivating your experiment.

State the research question.

Explain why an experimental design is appropriate.

Briefly describe your design approach (CRD, RBD, factorial, etc.).

End with a short paragraph outlining the paper structure.

2. Methods (Design and Execution)

Break this section into logical subsections:

2.1 Experimental Units, Treatments, and Blocking

Define the experimental unit precisely.

List all factors (treatment, blocking, nuisance) and their levels.

Explain the randomization procedure.

If applicable, show the alias structure or design matrix.

2.2 Randomization and Replication Plan

Describe the randomization process

State how replication was achieved and what it measures.

2.3 Execution and Data Collection

Describe how the experiment was conducted.

Report deviations from the plan and any missing data.

3. Statistical Analysis

Explain your model, contrasts, and diagnostics clearly.

3.1 Model Specification

Present the model formula (e.g., $Y \sim A * B + \text{Block}$).

State assumptions (independence, homoscedasticity, normality).

3.2 Planned Contrasts and Hypotheses

List primary hypotheses (e.g., main effects, interaction contrasts).

Clarify whether tests are one-sided or two-sided.

Report Type I error control (e.g., Tukey HSD, Fisher LSD).

3.3 Diagnostics

Include necessary residual plots and brief interpretations

3.4 Sensitivity / Robustness Analysis

Explore robustness to assumption violations (non-normality, heteroskedasticity).

Consider transformations, alternative models

4. Results

Present descriptive statistics (means, SDs, sample sizes) in a clear table.

Provide ANOVA or model summary tables.

Include plots illustrating estimated effects, confidence intervals, and interactions.

Emphasize interpretation in context: what do the effects mean practically?

Tip: Figures should complement—not duplicate—tables. Use clear axis labels and legends.

5. Discussion

Summarize key findings in plain language.

Interpret the results relative to the scientific question.

Discuss limitations: sample size, missing data, unmeasured confounders, etc.

Reflect on design efficiency, practical challenges, and lessons learned.

Suggest improvements for future experiments.

6. References

Use APA or journal style. Include citations for software packages (R or Python), design references (e.g., Montgomery, Box & Hunter), and any domain literature.

Appendices

Include all supplemental materials: codes, raw data, design matrix, randomization table, additional diagnostics, etc.

C. Evaluation Criteria (Manuscript Component)

Component	Points	Criteria
Abstract	2	Clear, quantitative summary
Introduction	3	Motivation, clarity
Methods	4	Proper structure, justification, rigor
Analysis & Diagnostics	3	Correct model, contrasts, diagnostics
Results	3	Clarity, relevance, and accuracy
Discussion	2	Insight, limitations, coherence
Reproducibility	2	Transparent code/data, consistent outputs
Others	1	Writing, references
Total	20	

D. Optional Enhancements for Distinction

Include a figure illustrating design efficiency. Clearly interpret what efficiency means in your context (variance, precision, or cost).

Perform a sequential design and demonstrate how the design adapts to accumulating data.

Compare alternative designs using collected data. Evaluate each design in terms of bias, precision, practical feasibility, and other factors.
